

RESEARCH PROGRESS UPDATE

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Experiment - 25 nodes - first half

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Raw NN Errors: [0.22558972 0.22677832 0.2487041 0.23937809 0.14248652 0.24240914
0.21907806 0.24059901 0.20481485 0.23738022 0.22263574 0.18443986
0.2091234 0.20350816 0.24351085 0.25063928 0.21539206 0.22985196
0.22254527 0.1602005 0.22594371 0.20306759 0.23631158 0.21247436
0.2129967 0.18102549 0.15307395 0.24566412 0.22484904 0.24645162
0.24797602 0.21822441 0.18379209 0.23700024 0.22630846 0.24236304
0.22872047 0.20101529 0.17038777 0.19500509 0.18410051 0.24294428
0.19977336 0.21679946 0.24427646 0.23477479 0.24443391 0.231328
0.23179669 0.21176254]
Raw KKThPINN Errors: [0.20664739 0.17516399 0.18977736 0.16492971 0.13628481 0.15734241
0.19086988 0.07548297 0.19998232 0.16222903 0.18272105 0.19406519
0.13684479 0.15685033 0.18385037 0.15110226 0.16597433 0.18907563
0.13551282 0.16351628 0.1979435 0.15582786 0.20414189 0.1526802
0.09494583 0.12660312 0.16661715 0.11553617 0.17058239 0.13242801
0.14712279 0.12858733 0.1669491 0.16370419 0.16320605 0.07712516
0.15480062 0.08997982 0.21134481 0.13691434 0.16129842 0.13217455
0.17233999 0.1888064 0.12849743 0.15842115 0.11847374 0.12690757
0.13525955 0.18535962]

=== Summary Statistics ===
NN Experiment Error - Mean: 0.2180741228, Variance: 0.0006920445
KKThPINN Experiment Error - Mean: 0.1556560333, Variance: 0.0010556299

=== Paired T-Test Results ===
T-statistic: 9.848470362207516
P-value: 3.3337516832162003e-13
Reject H0: KKThPINN and NN have significantly different performance (p = 0.0000)
KKThPINN performs significantly better.
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PRESENTATION TITLE

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Experiment – 30 nodes – first half

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Raw NN Errors: [0.20449958 0.19849467 0.18475673 0.22853967 0.2405842 0.19466579
0.16524643 0.19410593 0.21042929 0.22170514 0.17167205 0.17057568
0.18171109 0.24909002 0.12391764 0.22081026 0.21207834 0.21755727
0.22484275 0.22912537 0.18483456 0.12886086 0.19534104 0.18284286
0.17758408 0.19699909 0.14878697 0.1687904 0.19016915 0.14526946
0.19617801 0.16945797 0.21969876 0.20539538 0.15888601 0.17494986
0.19216592 0.13151375 0.24833824 0.20633434 0.18657278 0.20803412
0.15745019 0.130142 0.12955912 0.21775967 0.22399525 0.16677131
0.2149302 0.24054056]
Raw KKThPINN Errors: [0.11514162 0.16424346 0.09532966 0.11945478 0.16145152 0.10630153
0.20787257 0.17670841 0.15745449 0.16855825 0.08404593 0.13813208
0.13406736 0.11275236 0.07373782 0.16255262 0.16684414 0.13282213
0.12616143 0.18788377 0.17087972 0.17934165 0.09344246 0.18095943
0.0980373 0.14321642 0.17728682 0.07570955 0.0935602 0.18218567
0.16479971 0.07890727 0.15261459 0.16244407 0.14640046 0.19681543
0.1957745 0.16352357 0.07840797 0.14877364 0.0932382 0.19476056
0.1494454 0.14715366 0.17636367 0.10570066 0.14662758 0.1365855
0.18098189 0.09406027]

=== Summary Statistics ===
NN Experiment Error - Mean: 0.1908511957, Variance: 0.0010676191
KKThPINN Experiment Error - Mean: 0.1419902755, Variance: 0.0014127502

=== Paired T-Test Results ===
T-statistic: 6.62962690945028
P-value: 2.487111294613104e-08
Reject H0: KKThPINN and NN have significantly different performance (p = 0.0000)
KKThPINN performs significantly better.
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Training and Validation - 30 nodes - first half

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PS C:\Users\Fateme\Desktop\Research\CSTR\nonlinear\firsthalf\lessnodes\30\150> & C:/Users/Fateme/AppData/Local/Programs/Python/Python310/python.exe C:\Users\Fateme\Desktop\Research\CSTR\nonlinear\firsthalf\lessnodes\30\150> python C:\Users\Fateme\Desktop\Research\CSTR\nonlinear\firsthalf\lessnodes\30\150>
Raw NN Training Errors: [0.04186 0.04034 0.03318 0.05162 0.05792 0.0362 0.02689 0.03646 0.04424
0.04843 0.02868 0.02855 0.03234 0.0613 0.01496 0.0482 0.04433 0.04668
0.05019 0.04967 0.0338 0.01641 0.03772 0.03301 0.0321 0.0385 0.02147
0.02767 0.03573 0.02109 0.03803 0.02757 0.04834 0.04177 0.02497 0.03054
0.0367 0.01669 0.06052 0.04191 0.03488 0.04334 0.02347 0.01612 0.01655
0.0457 0.0507 0.02805 0.04501 0.05724]
Raw KThPINN Training Errors: [0.01293 0.02748 0.00868 0.01416 0.02587 0.01105 0.04166 0.03182 0.02476
0.02832 0.00677 0.01871 0.01749 0.01254 0.00525 0.02624 0.02687 0.01712
0.01541 0.03526 0.02889 0.03209 0.00854 0.03272 0.0089 0.02025 0.03146
0.00558 0.00803 0.03253 0.02632 0.00602 0.02381 0.02639 0.0208 0.03888
0.03684 0.02604 0.00602 0.02159 0.00854 0.03526 0.02203 0.02116 0.03078
0.011 0.02134 0.01842 0.03239 0.00858]
Raw NN Validation Errors: [0.04493 0.04338 0.03589 0.05696 0.06317 0.03996 0.02861 0.03998 0.0491
0.05381 0.031 0.03109 0.03498 0.06829 0.01544 0.05217 0.04814 0.05154
0.05509 0.05626 0.03658 0.0165 0.04103 0.03598 0.03459 0.04071 0.02264
0.03043 0.03787 0.02175 0.04168 0.03026 0.05271 0.04704 0.02606 0.03272
0.0393 0.01767 0.06763 0.04656 0.03791 0.04759 0.02546 0.01666 0.01712
0.05186 0.05558 0.02942 0.04995 0.06381]
Raw KThPINN Validation Errors: [0.01444 0.03015 0.01012 0.01604 0.02967 0.01242 0.04715 0.03501 0.02765
0.03173 0.00806 0.0211 0.01988 0.01416 0.0066 0.02977 0.0309 0.0192
0.0176 0.03962 0.03282 0.03644 0.00972 0.03596 0.01071 0.02331 0.03542
0.00685 0.00963 0.03756 0.02989 0.00737 0.02627 0.02972 0.02396 0.04398
0.04263 0.03007 0.00731 0.02481 0.00989 0.04208 0.02455 0.02391 0.03493
0.01223 0.02396 0.02032 0.03667 0.00994]

=== Summary Statistics ===
NN Training Error - Mean: 0.0369368000, Variance: 0.0001469868
KThPINN Training Error - Mean: 0.0211918000, Variance: 0.0001044726
NN Validation Error - Mean: 0.0402972000, Variance: 0.0001920565
KThPINN Validation Error - Mean: 0.0240836000, Variance: 0.0001324143

=== Paired T-Test Results ===
T-statistic: 6.647022690157631
P-value: 2.3374169382467674e-08
Reject H0: KThPINN and NN have significantly different performance (p = 0.0000)
KThPINN performs significantly better.

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Experiment - 32 nodes - first half

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Raw NN Errors: [0.14951394 0.22864499 0.2237136 0.16451419 0.21262258 0.20532982
0.20301673 0.18493566 0.22583773 0.13681823 0.14251868 0.2371089
0.20826546 0.18396073 0.17557593 0.17415026 0.21464598 0.18999087
0.13291057 0.22882981 0.20440237 0.1544534 0.08593772 0.18869485
0.17216135 0.20005607 0.24122932 0.22407501 0.18697143 0.17475581
0.15599003 0.19560421 0.19345923 0.1109514 0.21381398 0.16795515
0.15338152 0.12910703 0.15325593 0.21349715 0.11935037 0.19345954
0.12819359 0.14068072 0.22943375 0.21581593 0.13779546 0.16631878
0.20715735 0.1490349 ]
Raw KKThPINN Errors: [0.10493761 0.08057547 0.18757243 0.14101974 0.17388866 0.15533228
0.10595537 0.09143064 0.14194337 0.14805882 0.0725547 0.09947891
0.10995563 0.07295938 0.15068094 0.15702089 0.10125573 0.17389147
0.18485531 0.07454181 0.08963104 0.19881221 0.17262962 0.07383602
0.12133593 0.17552601 0.17009467 0.19968569 0.18197618 0.18109455
0.19376648 0.13579277 0.15774232 0.12953946 0.11443186 0.12761714
0.11981248 0.12995428 0.14764108 0.09848519 0.18818465 0.11463965
0.14187133 0.17856636 0.18076577 0.11481583 0.17922069 0.11358769
0.07371691 0.10216354]

=== Summary Statistics ===
NN Experiment Error - Mean: 0.1805979598, Variance: 0.0013707684
KKThPINN Experiment Error - Mean: 0.1366969308, Variance: 0.0015224140

=== Paired T-Test Results ===
T-statistic: 5.23951666714128
P-value: 3.3856163797601317e-06
Reject H0: KKThPINN and NN have significantly different performance (p = 0.0000)
KKThPINN performs significantly better.
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Training and Validation - 32 nodes - first half

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0.05095 0.0343 0.02992 0.02433 0.03819 0.0373 0.01141 0.04404 0.02755
0.02269 0.01658 0.02291 0.04373 0.0138 0.03683 0.01559 0.01959 0.05153
0.04589 0.01867 0.02759 0.0422 0.02178]
Raw KKThPINN Training Errors: [0.01079 0.00608 0.0345 0.02 0.0294 0.02388 0.01079 0.00811 0.01962
0.02152 0.00512 0.00937 0.01198 0.0051 0.02241 0.02332 0.01006 0.02987
0.03405 0.00513 0.00758 0.04074 0.02877 0.00528 0.01447 0.03127 0.028
0.03985 0.03286 0.03206 0.03783 0.01807 0.02451 0.0164 0.01283 0.016
0.01385 0.01648 0.02202 0.00962 0.03649 0.01273 0.02008 0.03135 0.03205
0.01314 0.0317 0.01236 0.005 0.01007]
Raw NN Validation Errors: [0.02341 0.05697 0.05437 0.02845 0.04876 0.04564 0.04452 0.03628 0.0556
0.01919 0.0208 0.06233 0.04731 0.03677 0.03288 0.032 0.05038 0.0389
0.0177 0.05639 0.04542 0.02461 0.00656 0.03835 0.03119 0.04327 0.06304
0.05605 0.03807 0.03167 0.0254 0.04175 0.04112 0.01182 0.0498 0.0292
0.02477 0.017 0.02428 0.04819 0.01398 0.04061 0.01637 0.02061 0.05685
0.05177 0.01948 0.02926 0.04642 0.02315]
Raw KKThPINN Validation Errors: [0.01216 0.00759 0.03879 0.02243 0.03374 0.02687 0.01261 0.0093 0.0223
0.02447 0.00649 0.01088 0.01335 0.00669 0.02499 0.02746 0.01156 0.03416
0.03811 0.00668 0.00879 0.04554 0.03262 0.00655 0.01612 0.0349 0.03239
0.04425 0.03644 0.03716 0.04201 0.02032 0.02737 0.0186 0.01459 0.01811
0.01552 0.01841 0.02392 0.0108 0.0407 0.01415 0.02251 0.03537 0.03683
0.0148 0.03564 0.0143 0.00653 0.01163]

=== Summary Statistics ===
NN Training Error - Mean: 0.0334928000, Variance: 0.0001655796
KKThPINN Training Error - Mean: 0.0198912000, Variance: 0.0001148198
NN Validation Error - Mean: 0.0363742000, Variance: 0.0002146524
KKThPINN Validation Error - Mean: 0.0225500000, Variance: 0.0001413538

=== Paired T-Test Results ===
T-statistic: 5.285619510078781
P-value: 2.88480189245347e-06
Reject H0: KKThPINN and NN have significantly different performance (p = 0.0000)
KKThPINN performs significantly better.
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Experiment - 35 nodes - first half

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Raw NN Errors: [0.214686  0.12500288 0.21951202 0.08896682 0.14359744 0.11098481
0.2198008 0.12295097 0.21553426 0.07375548 0.1940851 0.21436608
0.18392162 0.17034658 0.1063947 0.21436292 0.23431272 0.21371983
0.22806138 0.22561002 0.18795703 0.1476282 0.20362008 0.23113894
0.23337604 0.22627945 0.11732778 0.13029634 0.18779999 0.21049207
0.14655312 0.10153045 0.21083322 0.10071921 0.1819483 0.19761151
0.21259781 0.22733925 0.16046298 0.22457659 0.20887976 0.17598184
0.17748625 0.06579536 0.06586568 0.20363783 0.16919786 0.17739159
0.2381201 0.15256746]
Raw KKThPINN Errors: [0.08985873 0.12723887 0.20062426 0.17153304 0.10345268 0.16092278
0.10266859 0.14888099 0.10534355 0.17963194 0.08655427 0.10621035
0.14437994 0.07045712 0.14745287 0.10035042 0.07884799 0.12060928
0.10363848 0.16210627 0.1877495 0.08092928 0.10720635 0.13042763
0.12586084 0.08588805 0.07418105 0.15292628 0.16013201 0.07606268
0.11582919 0.13658852 0.14429973 0.18308912 0.07173012 0.13859623
0.13652606 0.11043722 0.15221398 0.11634517 0.1327051 0.12418515
0.14385708 0.15682893 0.09174233 0.12626485 0.11706706 0.11444116
0.08530475 0.08002327]

=== Summary Statistics ===
NN Experiment Error - Mean: 0.1758996903, Variance: 0.0024934236
KKThPINN Experiment Error - Mean: 0.1234040220, Variance: 0.0011157749

=== Paired T-Test Results ===
T-statistic: 5.4763024060868295
P-value: 1.4833650267118391e-06
Reject H0: KKThPINN and NN have significantly different performance (p = 0.0000)
KKThPINN performs significantly better.
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Training and Validation - 35 nodes - first half

```
0.0104 0.02609 0.03471 0.00657 0.01091 0.01628 0.01576 0.00726 0.00519
0.02343 0.02436 0.0059 0.01295 0.0185 0.02067 0.03146 0.00505 0.01888
0.01822 0.01172 0.02313 0.01331 0.0176 0.01557 0.02063 0.02465 0.00807
0.01612 0.01353 0.01255 0.00715 0.00619]
Raw NN Validation Errors: [0.04973 0.01534 0.05261 0.00715 0.02128 0.01184 0.05287 0.01483 0.05188
0.00486 0.04062 0.05106 0.03666 0.03064 0.0109 0.05 0.05968 0.05021
0.05823 0.05461 0.03755 0.02225 0.04514 0.05873 0.0586 0.05721 0.01332
0.0168 0.03743 0.04823 0.02215 0.00975 0.0494 0.00956 0.03549 0.04208
0.04897 0.05647 0.02669 0.05566 0.04805 0.03265 0.03341 0.00355 0.00353
0.04462 0.03051 0.03334 0.06302 0.02461]
Raw KThPINN Validation Errors: [0.00937 0.01792 0.04586 0.03271 0.01165 0.02863 0.01163 0.02473 0.0119
0.03572 0.0085 0.01249 0.02287 0.00632 0.02366 0.01127 0.00697 0.01587
0.012 0.02957 0.03926 0.00765 0.01258 0.01884 0.01755 0.00854 0.00672
0.02593 0.02877 0.00693 0.01473 0.02076 0.02362 0.03729 0.00634 0.02066
0.02033 0.01363 0.02623 0.01514 0.01911 0.01705 0.02292 0.02718 0.00948
0.01757 0.01534 0.01435 0.0083 0.00749]

=== Summary Statistics ===
NN Training Error - Mean: 0.0329920000, Variance: 0.0002551118
KThPINN Training Error - Mean: 0.0160802000, Variance: 0.0000747579
NN Validation Error - Mean: 0.0358754000, Variance: 0.0003279628
KThPINN Validation Error - Mean: 0.0181986000, Variance: 0.0000916524

=== Paired T-Test Results ===
T-statistic: 5.882182765145983
P-value: 3.552920981382262e-07
Reject H0: KThPINN and NN have significantly different performance (p = 0.0000)
KThPINN performs significantly better.
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